

SARNIA CHRIS HADFIELD, ON, Canada

WMO: 717040

Lat: 42.9950N

Lon: 82.3075W

Elev: 181

StdP: 99.16

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.4	-14.6	-22.0	0.5	-16.8	-19.2	0.7	-14.2	12.7	-5.5	11.6	-5.2	4.0	310	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	31.1	22.5	29.6	21.8	28.1	21.1	24.1	28.9	23.1	27.5	22.3	26.3	4.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.8	26.6	21.8	16.8	25.4	21.0	16.0	24.8	73.5	28.9	69.7	27.5	66.5	26.4	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.5	8.4	DB	-20.7	34.2	3.9	1.3	-23.5	35.1	-25.8	35.9	-27.9	36.7	-30.8	37.6	(4)
(5)			WB	-21.1	26.2	3.8	1.3	-23.8	27.1	-26.0	27.9	-28.1	28.6	-30.8	29.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-4.0	-4.0	1.0	7.2	13.8	18.7	21.7	20.9	17.5	11.3	4.4	-0.8	(6)	
	DBStd	10.33	5.88	5.97	5.78	4.85	5.08	3.84	3.08	2.86	3.84	4.60	4.86	4.72	(7)	
	HDD10.0	1794	433	393	287	113	17	0	0	0	0	38	177	335	(8)	
	HDD18.3	3701	691	626	539	334	159	42	6	8	61	224	417	593	(9)	
	CDD10.0	1444	1	0	7	29	136	260	362	338	224	79	9	0	(10)	
	CDD18.3	310	0	0	0	0	19	52	109	88	36	6	0	0	(11)	
	CDH23.3	2355	0	0	3	13	211	441	836	563	247	41	0	0	(12)	
	CDH26.7	661	0	0	0	1	55	135	253	141	67	8	0	0	(13)	
(14) Wind	WSAvg	4.1	4.7	4.6	4.7	4.8	4.0	3.6	3.2	3.1	3.4	4.2	4.5	4.5	(14)	
(15) Precipitation	PrecAvg	866	48	52	67	75	68	80	69	77	92	64	86	73	(15)	
	PrecMax	1102	85	109	146	132	123	135	108	180	251	119	134	117	(16)	
	PrecMin	640	24	15	30	39	23	21	31	21	12	17	17	33	(17)	
	PrecStd	115	20	29	27	23	27	27	25	50	61	29	35	21	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	13.8	22.0	25.2	30.3	32.4	33.4	32.3	31.2	27.3	19.4	13.7	(19)	
		MCWB	10.9	10.8	16.6	16.5	20.7	22.7	24.1	23.3	22.8	21.2	14.1	12.2	(20)	
	2%	DB	8.5	9.4	16.1	21.9	28.4	30.3	31.2	29.9	28.6	23.8	16.0	10.2	(21)	
		MCWB	7.3	7.5	12.1	14.8	19.9	21.5	23.0	22.2	21.8	18.3	12.6	8.7	(22)	
	5%	DB	5.9	6.0	12.6	18.8	25.8	28.1	29.5	28.2	26.2	20.9	13.8	7.6	(23)	
		MCWB	4.4	4.2	9.6	12.9	18.6	20.7	22.1	21.7	20.7	16.8	10.7	6.0	(24)	
	10%	DB	3.8	3.2	9.4	15.8	23.0	25.8	27.7	26.4	23.9	18.6	11.7	5.2	(25)	
		MCWB	2.8	1.6	6.8	11.2	16.7	19.5	21.2	20.8	19.1	15.0	9.3	3.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	11.4	16.7	17.7	22.7	24.4	25.6	25.4	24.4	22.1	15.1	12.2	(27)	
		MCDB	11.8	13.0	21.6	22.8	28.9	30.2	30.8	28.8	29.5	27.2	17.0	13.5	(28)	
	2%	WB	7.6	7.6	12.7	15.7	21.0	22.7	24.2	23.7	22.7	19.5	13.0	9.0	(29)	
		MCDB	8.3	9.3	14.7	20.2	26.2	27.7	29.1	27.9	26.8	22.2	15.5	10.0	(30)	
	5%	WB	4.7	4.0	9.9	13.7	19.4	21.7	23.3	22.6	21.5	17.3	11.1	6.2	(31)	
		MCDB	5.5	5.5	12.3	18.1	24.3	26.3	27.5	26.3	24.8	20.2	13.1	7.5	(32)	
	10%	WB	2.8	1.8	6.8	11.5	17.9	20.5	22.4	21.8	20.2	15.5	9.3	4.0	(33)	
		MCDB	3.8	3.1	8.9	15.2	22.0	24.3	26.2	25.1	22.9	18.2	11.4	5.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	7.5	8.1	9.7	10.6	9.7	9.3	9.0	9.5	8.4	7.3	5.7	(35)	
		MCDBR	8.1	10.8	13.2	15.6	14.2	12.9	11.9	11.3	12.1	11.9	10.8	8.7	(36)	
	5% WB	MCWBR	7.9	9.5	10.0	9.8	8.0	6.6	5.7	5.7	6.3	7.1	8.2	8.1	(37)	
		MCWBR	7.9	10.3	12.7	13.8	12.9	11.0	10.0	9.3	10.2	10.4	9.7	8.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.309	0.320	0.339	0.382	0.407	0.431	0.418	0.408	0.376	0.359	0.319	0.294	(40)	
		taud	2.341	2.303	2.368	2.277	2.263	2.222	2.286	2.302	2.380	2.403	2.485	2.446	(41)	
	Ebn at Noon		818	872	903	883	868	844	851	848	852	820	810	811	(42)	
		Edh at Noon	82	101	107	127	133	139	129	123	106	91	72	68	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.33	2.14	3.46	4.56	5.38	6.10	6.01	5.24	4.19	2.65	1.59	1.08	(44)	
	RadStd		0.17	0.23	0.36	0.40	0.46	0.35	0.33	0.24	0.37	0.23	0.21	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page